



AI Anti-Fraud 3.0 Components

Api Proxy
Exposes http://localhost:9000 on the host.
Acts as the public entry point for the system.
Reverse proxies requests to the app service and load balances across scaled app instances.

App
Flask API service that exposes /api/fraud.
Accepts a fraud-investigation question from the user.
Sends chat messages to the model service to obtain a tool call and a final response.
Executes the generated SQL against the SQLite database.
Can be scaled horizontally, for example with --scale app=3.

Model
Separate Flask service that exposes /generate and /health.
Loads the Apertus base model together with the pwned-next adapter.
Performs inference for the app service.
Runs as a single shared inference backend for all app instances.

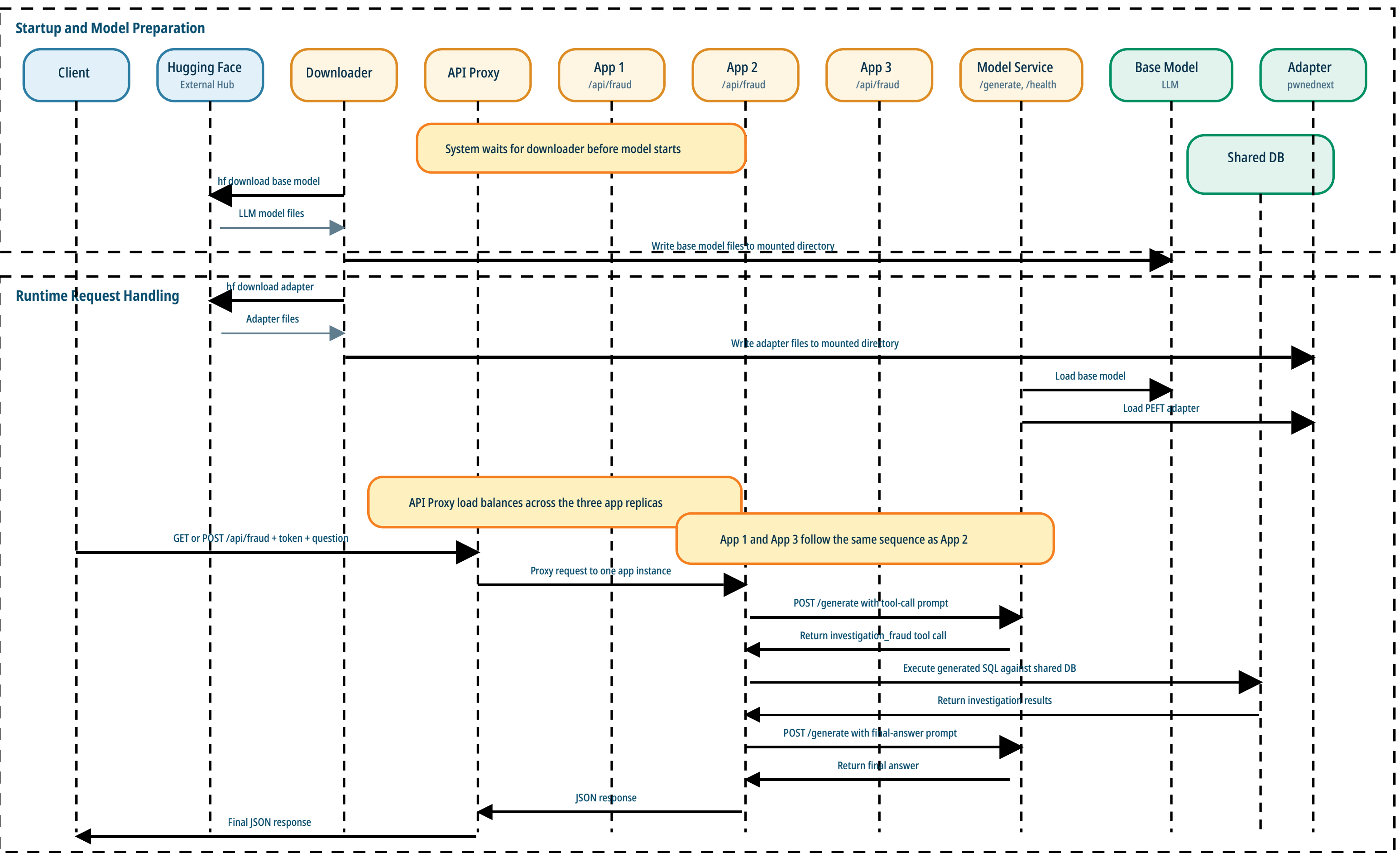
Downloader
One-shot setup container used during startup.
Downloads the base model and adapter from Hugging Face if they are not already present.
Writes those artifacts into shared mounted directories used by the model service.

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The AI Anti-Fraud 3.0 is deployed as a small microservice system. It separates request handling, model inference, and supporting services so the application can be scaled and threat-modeled more easily.

Sequence Diagram

Startup download flow and runtime request flow across the system deployment



DFD Diagram

